

Notes on Measure Theory: Absolute Continuity and the Radon–Nikodym Theorem

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August 13, 2026

Abstract

Results on how one measure sits inside another: absolute continuity and mutual singularity, the Jordan decomposition of a signed measure, the Radon–Nikodym theorem with its calculus (linearity, the chain rule, and change of variables), and the Lebesgue decomposition theorem. The running theme is that a density $\frac{d\nu}{d\mu}$ is the measure-theoretic object that both probability densities and conditional expectations instantiate.

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1 Absolute continuity and singularity

Throughout, $(X, \mathcal{A})$ is a measurable space and all measures are defined on $\mathcal{A}$. A measure μ is σ -finite if $X = \bigcup_n A_n$ with $A_n \in \mathcal{A}$ and $\mu(A_n) < \infty$; σ -finiteness is the standing hypothesis under which the theorems below hold in full generality.

Definition 1.1 (Absolute continuity). Let μ, ν be measures on $(X, \mathcal{A})$. We say ν is *absolutely continuous* with respect to μ , written $\nu \ll \mu$, if

$$\mu(A) = 0 \implies \nu(A) = 0 \quad \text{for every } A \in \mathcal{A}. \quad (1)$$

Equivalently: every μ -null set is ν -null.

Definition 1.2 (Mutual singularity). The measures μ, ν are *mutually singular*, written $\mu \perp \nu$, if there is a set $E \in \mathcal{A}$ with

$$\mu(E) = 0 \quad \text{and} \quad \nu(X \setminus E) = 0, \quad (2)$$

i.e. they live on disjoint sets: ν is carried by E and μ by its complement.

Absolute continuity and singularity are the two extremes. The Lebesgue decomposition of Section 3 shows every σ -finite ν splits uniquely into a piece of each type relative to μ .

The ε - δ form. For *finite* ν , the set-theoretic Definition 1.1 is equivalent to an analytic continuity statement, which is where the name comes from.

Proposition 1.3 (ε - δ characterization). *Let ν be a finite measure. Then $\nu \ll \mu$ if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$A \in \mathcal{A}, \mu(A) < \delta \implies \nu(A) < \varepsilon. \quad (3)$$

Proof sketch. The ε - δ condition trivially forces $\mu(A) = 0 \implies \nu(A) = 0$. Conversely, if (3) failed there would be $\varepsilon_0 > 0$ and sets A_n with $\mu(A_n) < 2^{-n}$ but $\nu(A_n) \geq \varepsilon_0$. Put $A = \limsup_n A_n$. By Borel–Cantelli on μ , $\mu(A) = 0$, while $\nu(A) \geq \varepsilon_0$ by finiteness of ν and reverse Fatou, contradicting $\nu \ll \mu$. $\square$

Remark. Finiteness matters in Proposition 1.3. With μ = Lebesgue measure and $\nu(A) = \int_A x^2 dx$ on $\mathbb{R}$, one has $\nu \ll \mu$ (Definition 1.1) yet no single δ works uniformly for the interval family $\{[n, n + \delta]\}$, because ν is not finite. The remedy is to test (3) on sets of finite ν -measure.

Signed measures and the Jordan decomposition. A *signed measure* ν assigns values in $(-\infty, +\infty]$ (or $[-\infty, +\infty)$) and is countably additive. The Hahn decomposition gives a partition $X = P \sqcup N$ into a positive set P and a negative set N ; setting $\nu^+(A) = \nu(A \cap P)$ and $\nu^-(A) = -\nu(A \cap N)$ yields two genuine (nonnegative) measures.

Proposition 1.4 (Jordan decomposition). *Every signed measure ν admits a unique decomposition*

$$\nu = \nu^+ - \nu^-, \quad \nu^+ \perp \nu^-, \quad (4)$$

into mutually singular measures. Its total variation is $|\nu| = \nu^+ + \nu^-$, and $\nu \ll \mu$ is defined to mean $|\nu| \ll \mu$.

This reduces every statement about signed measures to the nonnegative case applied to ν^+ and ν^- separately, which is how the Radon–Nikodym theorem is extended to signed and complex measures.

2 The Radon–Nikodym theorem

The theorem says that absolute continuity is not merely a null-set condition: whenever $\nu \ll \mu$, the measure ν is obtained from μ by integrating a density. It is the converse to the elementary fact that $A \mapsto \int_A f d\mu$ defines a measure absolutely continuous with respect to μ .

Theorem 2.1 (Radon–Nikodym). *Let μ and ν be σ -finite measures on $(X, \mathcal{A})$ with $\nu \ll \mu$. Then there exists a measurable function $f : X \rightarrow [0, \infty)$ such that*

$$\nu(A) = \int_A f d\mu \quad \text{for all } A \in \mathcal{A}. \quad (5)$$

The function f is unique up to μ -a.e. equality. It is called the Radon–Nikodym derivative of ν with respect to μ and written

$$f = \frac{d\nu}{d\mu}. \quad (6)$$

Remark (Why σ -finiteness is needed). Take μ the counting measure and ν the Lebesgue measure on $[0, 1]$. Then $\nu \ll \mu$ (the only μ -null set is $\emptyset$), yet no f satisfies (5): it would need $\int_{\{x\}} f d\mu = f(x) = \nu(\{x\}) = 0$ for every x , forcing $f \equiv 0$ and hence $\nu \equiv 0$. Counting measure is not σ -finite on an uncountable space, and the theorem fails.

Remark (The von Neumann proof). The slick proof avoids constructing f by hand. Assume μ, ν finite, set $\rho = \mu + \nu$, and consider the bounded linear functional $g \mapsto \int g d\nu$ on the Hilbert space $L^2(\rho)$. By Riesz representation it equals $\int gh d\rho$ for some $h \in L^2(\rho)$; algebra on (5)-type identities shows $0 \leq h < 1$ μ -a.e. (this uses $\nu \ll \mu$), and $f = h/(1-h)$ is the required derivative. The σ -finite case follows by exhausting X with finite pieces.

2.1 The calculus of Radon–Nikodym derivatives

The derivative notation is justified by the following rules, each holding μ - (respectively λ -) almost everywhere. They are what make $\frac{d\nu}{d\mu}$ behave like an ordinary derivative.

Proposition 2.2 (Linearity). *If $\nu_1, \nu_2 \ll \mu$ are σ -finite, then $\nu_1 + \nu_2 \ll \mu$ and*

$$\frac{d(\nu_1 + \nu_2)}{d\mu} = \frac{d\nu_1}{d\mu} + \frac{d\nu_2}{d\mu} \quad \mu\text{-a.e.} \quad (7)$$

Proposition 2.3 (Change of variables). *Let $\nu \ll \mu$ be σ -finite with density $f = \frac{d\nu}{d\mu}$. For every measurable $g : X \rightarrow [0, \infty]$,*

$$\int_X g d\nu = \int_X g f d\mu = \int_X g \frac{d\nu}{d\mu} d\mu. \quad (8)$$

The identity extends to ν -integrable g of arbitrary sign.

Proof idea. Equation (5) is exactly (8) for indicators $g = \mathbf{1}_A$. Extend by linearity to simple functions, then by monotone convergence to nonnegative measurable g , then split $g = g^+ - g^-$ for the signed case. $\square$

Proposition 2.4 (Chain rule). *If $\nu \ll \mu \ll \lambda$ with all three σ -finite, then $\nu \ll \lambda$ and*

$$\frac{d\nu}{d\lambda} = \frac{d\nu}{d\mu} \frac{d\mu}{d\lambda} \quad \lambda\text{-a.e.} \quad (9)$$

In particular, if $\mu \ll \nu$ and $\nu \ll \mu$ (equivalent measures) then $\frac{d\nu}{d\mu}$ is strictly positive μ -a.e. and

$$\frac{d\mu}{d\nu} = \left(\frac{d\nu}{d\mu} \right)^{-1} \quad \mu\text{-a.e.} \quad (10)$$

Proof idea. Apply Proposition 2.3 with $g = \mathbf{1}_A \frac{d\mu}{d\nu}$ and the density $\frac{d\mu}{d\lambda}$: $\nu(A) = \int_A \frac{d\nu}{d\mu} d\mu = \int_A \frac{d\nu}{d\mu} \frac{d\mu}{d\lambda} d\lambda$, so the product is a λ -density for ν ; uniqueness in Theorem 2.1 gives (9). The reciprocal rule is the case $\lambda = \nu$, where $\frac{d\nu}{d\nu} = 1$. $\square$

3 The Lebesgue decomposition and two instances

Radon–Nikodym handles the absolutely continuous case. The Lebesgue decomposition theorem says that this case, together with the singular one, is all there is: any σ -finite measure splits uniquely into an $\frac{d\nu}{d\mu}$ -representable part and a part concentrated where μ is not.

Theorem 3.1 (Lebesgue decomposition). *Let μ and ν be σ -finite measures on $(X, \mathcal{A})$. There is a unique pair of σ -finite measures $\nu_{\text{ac}}, \nu_{\text{s}}$ with*

$$\nu = \nu_{\text{ac}} + \nu_{\text{s}}, \quad \nu_{\text{ac}} \ll \mu, \quad \nu_{\text{s}} \perp \mu. \quad (11)$$

Consequently ν_{ac} has a density $\frac{d\nu_{\text{ac}}}{d\mu}$, and one writes $\frac{d\nu}{d\mu} := \frac{d\nu_{\text{ac}}}{d\mu}$ for the Radon–Nikodym part even when $\nu \not\ll \mu$.

Proof idea. Apply Theorem 2.1 to ν against the reference $\rho = \mu + \nu$: $\nu \ll \rho$, so $\nu = \int h d\rho$ for some $0 \leq h \leq 1$. The set $E = \{h = 1\}$ carries the singular part $\nu_{\text{s}} = \nu|_E$, its complement the absolutely continuous part. Uniqueness follows because a measure that is both $\ll \mu$ and $\perp \mu$ must vanish. $\square$

Corollary 3.2. *$\nu \ll \mu$ if and only if $\nu_{\text{s}} = 0$; and $\nu \perp \mu$ if and only if $\nu_{\text{ac}} = 0$. The two notions of Section 1 are exactly the vanishing of one summand.*

Instance 1: probability densities. Let $\mathbb{P}$ be the law of a real random variable X and let μ be Lebesgue measure on $\mathbb{R}$. The distribution decomposes (Theorem 3.1) into an absolutely continuous part with a probability density function $p = \frac{d\mathbb{P}_{\text{ac}}}{d\mu}$, an atomic part (a sum of point masses, singular w.r.t. μ), and a singular continuous part (e.g. the Cantor distribution). “ X has density p ” is precisely the statement $\mathbb{P} \ll \mu$ with $\frac{d\mathbb{P}}{d\mu} = p$; then, by Proposition 2.3,

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g d\mathbb{P} = \int_{\mathbb{R}} g(x) p(x) dx. \quad (12)$$

This is the exact sense in which the likelihood notes’ “density” is a Radon–Nikodym derivative: there $f(x | \theta) = \frac{dP_{\theta}}{d\mu}$.

Instance 2: conditional expectation. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space, $\mathcal{G} \subseteq \mathcal{A}$ a sub- σ -algebra, and $X \geq 0$ integrable. Define a measure on $\mathcal{G}$ by

$$\nu(G) = \int_G X d\mathbb{P}, \quad G \in \mathcal{G}. \quad (13)$$

Then $\nu \ll \mathbb{P}|_{\mathcal{G}}$ (a $\mathbb{P}$ -null set in $\mathcal{G}$ is ν -null), so Radon–Nikodym supplies a $\mathcal{G}$ -measurable density.

Definition 3.3 (Conditional expectation). The *conditional expectation* of X given $\mathcal{G}$ is

$$\mathbb{E}[X | \mathcal{G}] = \frac{d\nu}{d\mathbb{P}|_{\mathcal{G}}}, \quad (14)$$

the Radon–Nikodym derivative on $\mathcal{G}$. It is the unique (up to $\mathbb{P}$ -a.s.) $\mathcal{G}$ -measurable random variable with $\int_G \mathbb{E}[X | \mathcal{G}] d\mathbb{P} = \int_G X d\mathbb{P}$ for all $G \in \mathcal{G}$.

Existence and a.s.-uniqueness of conditional expectation are therefore not separate theorems: they are Theorem 2.1 read on the sub- σ -algebra $\mathcal{G}$. The defining averaging identity is the special case $A = G$ of (5).